

Here is a professional, production-ready deliverable format specifically designed for **Project 1 (Active Directory Architecture & Deployment)**.

Since an Active Directory project focuses on access control, organizational structure, and identity management, your repository and documentation need to emphasize **hierarchical design**, **automation scripts**, and **remediation proof**.

Portfolio Folder View

Structure this project folder inside your repository to show you can organize enterprise configurations:

Plaintext

-  security-homelab/
- └─  01-active-directory-deployment/
- │ └─ README.md <-- (The documentation layout below)
- │ └─  documentation/
- │ └─ AD-Design-Spec.pdf <-- Optional: A deeper design document or PDF layout
- │ └─  scripts/
- │ └─ Create-OUTs.ps1 <-- Your PowerShell script that builds the OUTs
- │ └─ Populate-Users.ps1 <-- Your PowerShell script that injects mock users

Markdown Deliverable Template (README.md)

Use this exact structural blueprint to write out the final documentation for your Active Directory project:

Markdown

- [illegible]

- [Mock OU: IT_Admns Group]
- [Mock OU: HR_Staff Group]
- [Mock OU: Finance_Users Group]

Environment Specs:

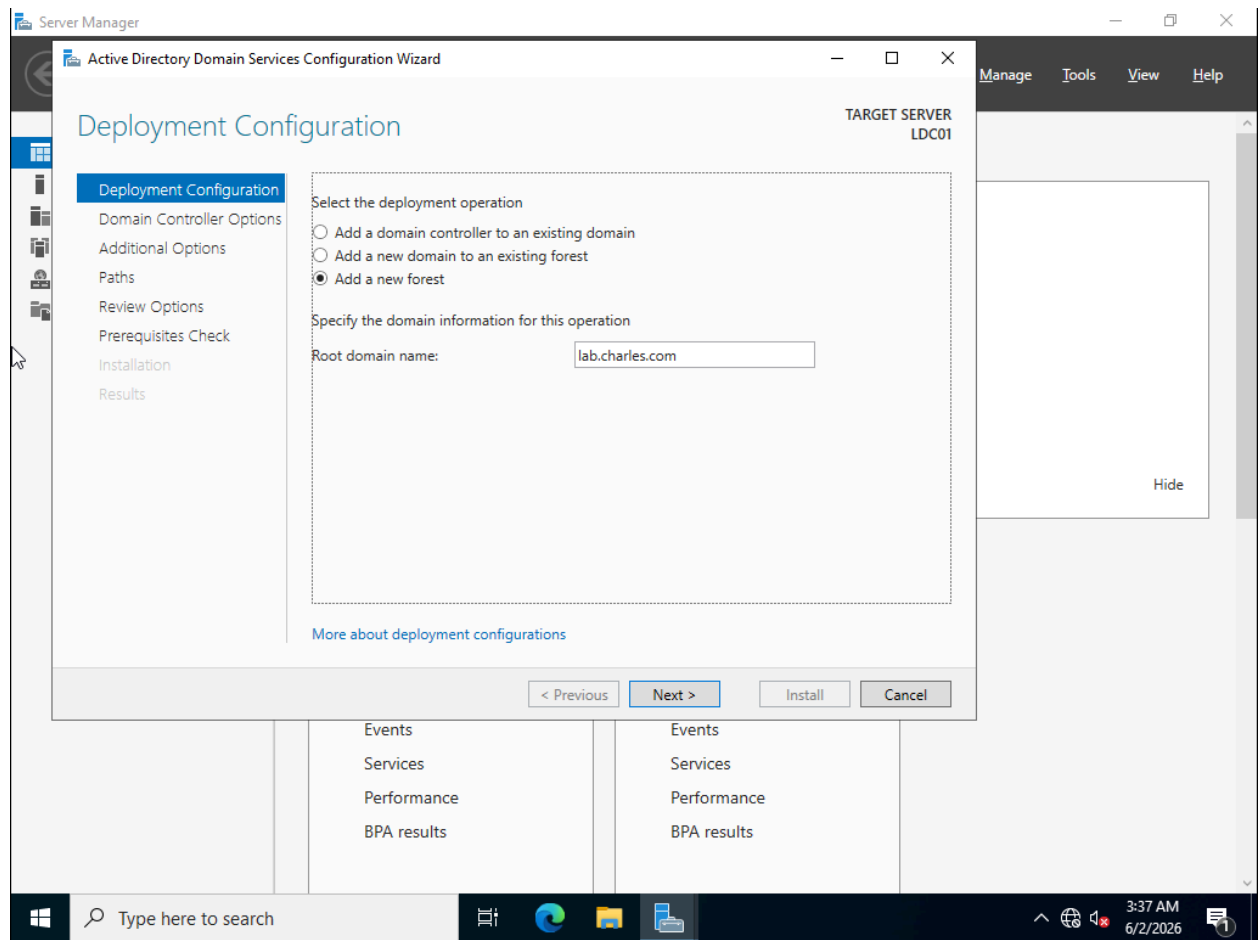
- **Operating System:** Windows Server 2022 Standard (Desktop Experience)
- **Domain Name:** lab.charles.com
- **Network Configuration:** Dedicated Host-Only Segment / LAN Segment (192.168.10.0/24)

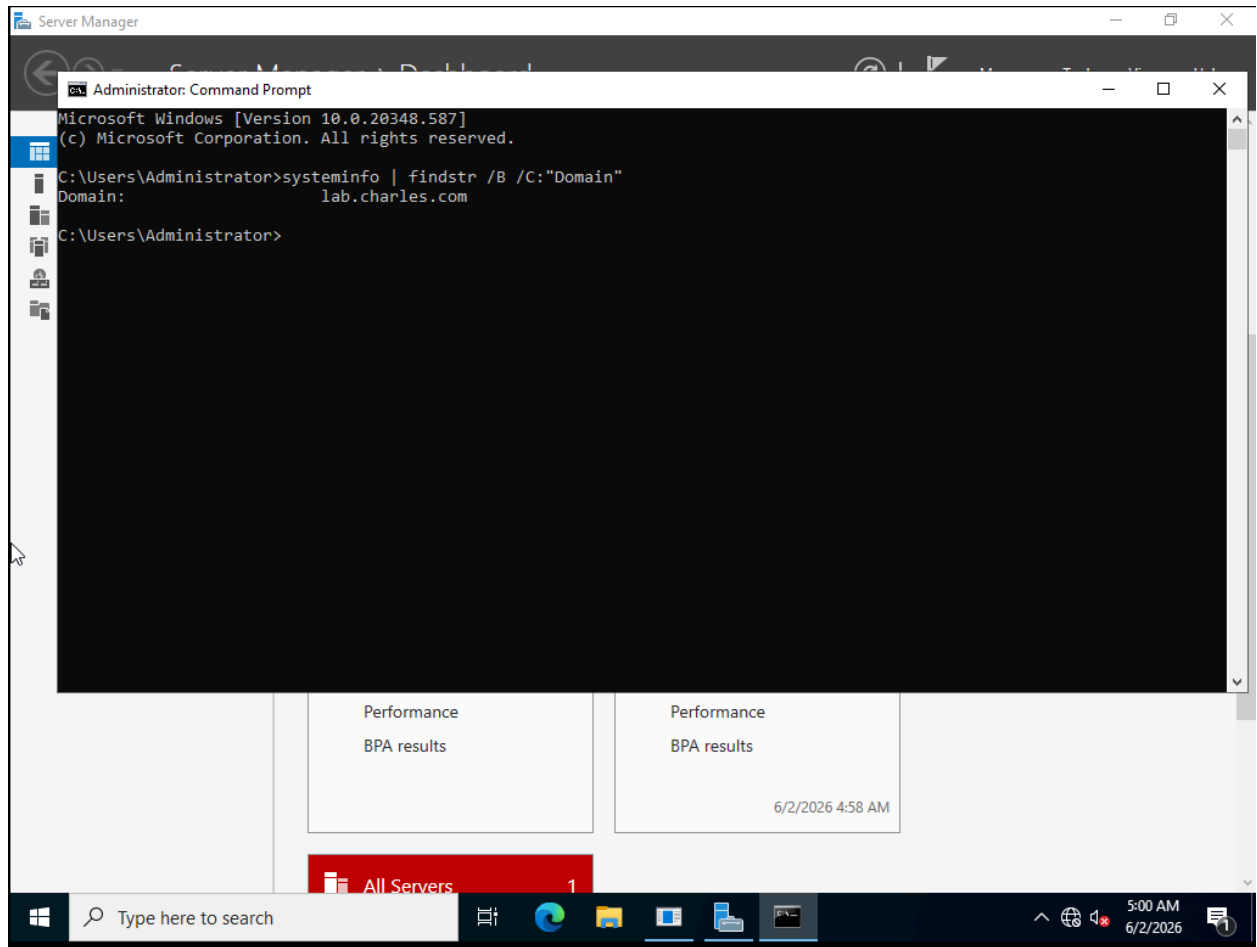
◆ Step-by-Step Implementation

1. Network Core Configuration & Role Promotion

Before installing Active Directory, the base server operating system was provisioned with a static IP configuration to prevent DNS resolution failures.

- Assigned a static IPv4 address (192.168.10.10) and pointed the primary DNS server to the loopback address (127.0.0.1).
- Installed the **Active Directory Domain Services (AD DS)** and **DNS Server** roles using Server Manager.
- Promoted the server to a Domain Controller (DC), establishing a brand-new forest named lab.charles.com.



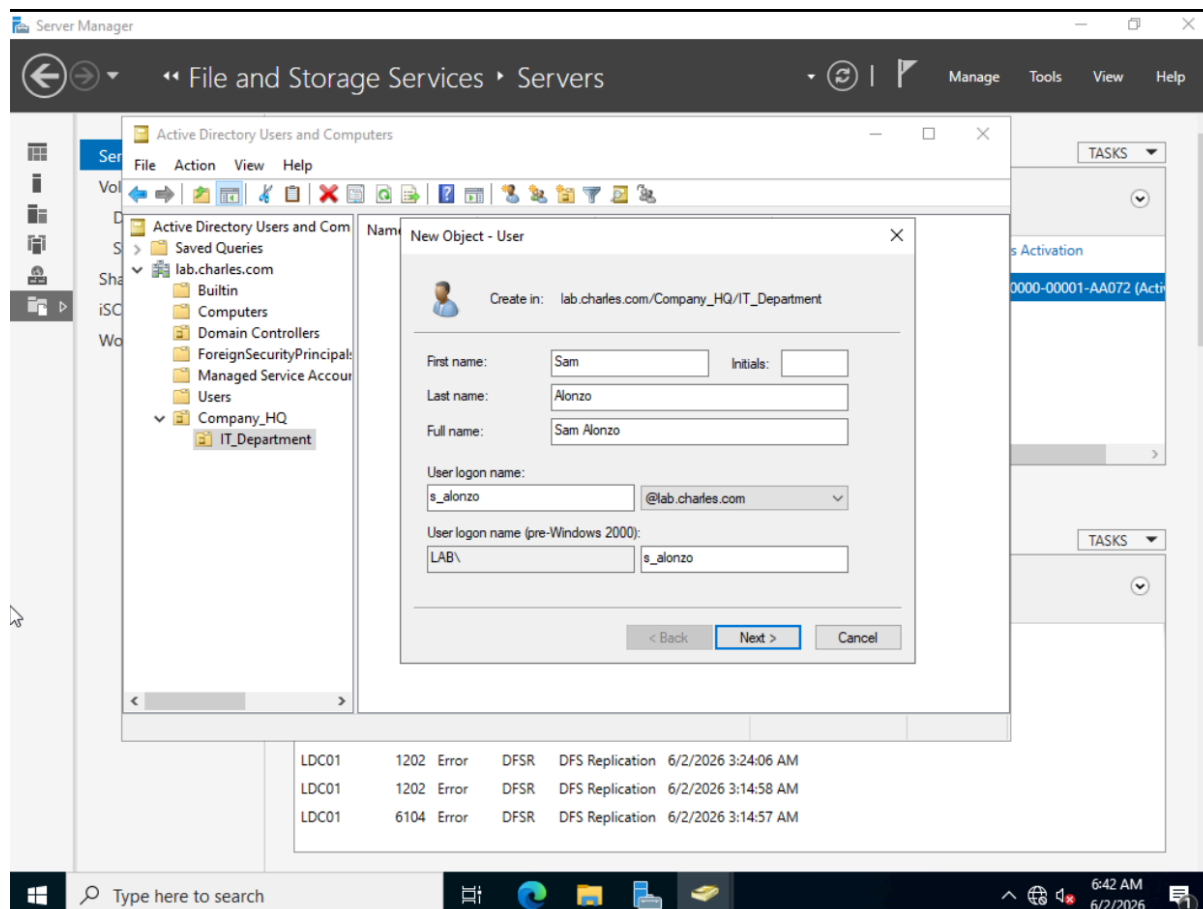
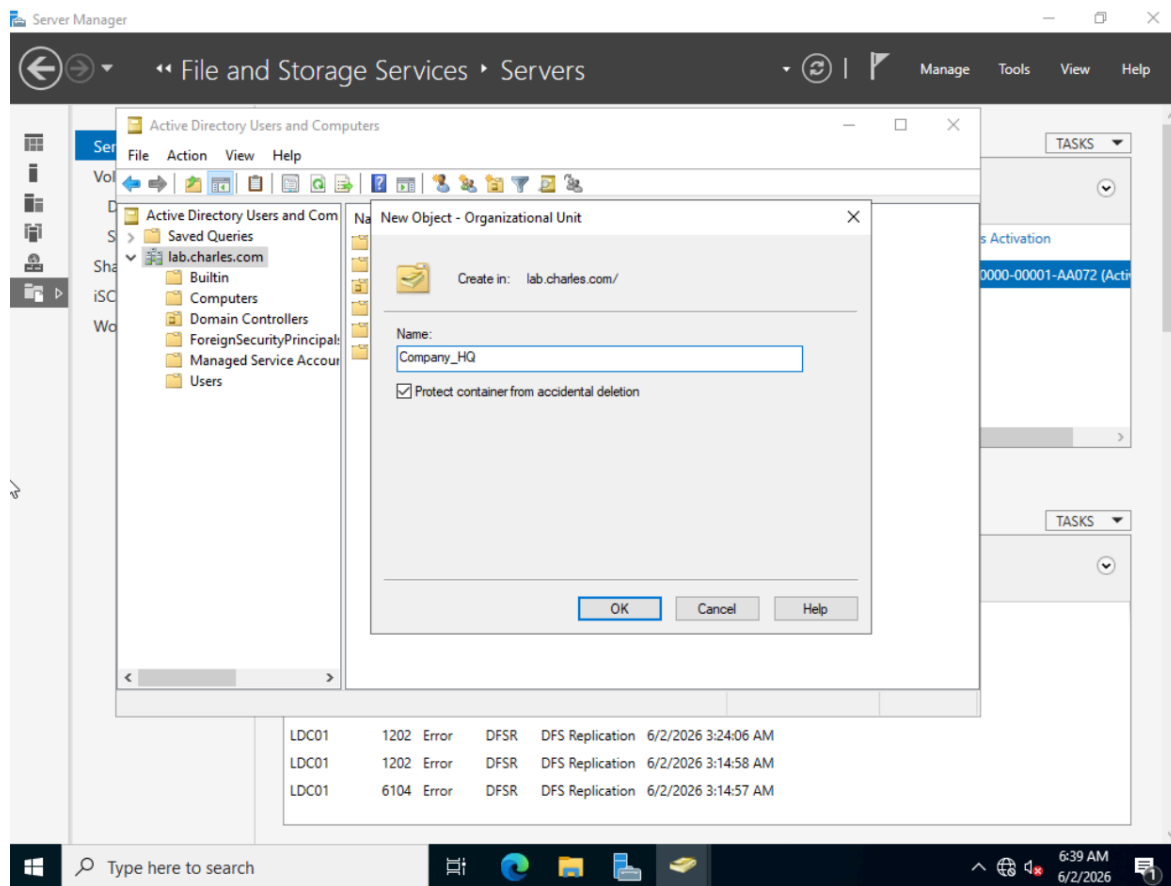


2. Automated Logical Architecture Design (OUs and Groups)

To demonstrate production-level scalability, the entire corporate hierarchy was deployed programmatically via PowerShell scripts rather than clicking through the GUI manually.

- **Organizational Units (OUs) Created:** [Corporate_HQ](#), [Departments](#), [Users](#), [Groups](#), and [Service_Accounts](#).
- **Security Groups Provisioned:** Created distinct functional roles ([GG_IT_Admins](#), [GG_HR](#), [GG_Finance](#)) to enforce the Principle of Least Privilege.

OU/User Adding Using GUI



Server Manager

File and Storage Services > Servers

Active Directory Users and Computers

File Action View Help

lab.charles.com

- Builtin
- Computers
- Domain Controllers
- ForeignSecurityPrincipals
- Managed Service Accounts
- Users
- Company_HQ
 - IT_Department

Name	Type	Description
Sam Alonzo	User	

Tasks

Tasks

Source	Destination	Result	Source	Destination	Result
LDC01	1202	Error	DFSR	DFS Replication	6/2/2026 3:24:06 AM
LDC01	1202	Error	DFSR	DFS Replication	6/2/2026 3:14:58 AM
LDC01	6104	Error	DFSR	DFS Replication	6/2/2026 3:14:57 AM

Type here to search

6:48 AM 6/2/2026

Server Manager

File and Storage Services > Servers

Active Directory Users and Computers

File Action View Help

lab.charles.com

- Builtin
- Computers
- Domain Controllers
- ForeignSecurityPrincipals
- Managed Service Accounts
- Users
- Company_HQ
 - IT_Department

New Object - Group

Create in: lab.charles.com/Company_HQ/IT_Department

Group name: GG_IT_Admins

Group name (pre-Windows 2000): GG_IT_Admins

Group scope: ☒ Domain local ☐ Global ☐ Universal

Group type: ☒ Security ☐ Distribution

OK Cancel

Source	Destination	Result	Source	Destination	Result
LDC01	1202	Error	DFSR	DFS Replication	6/2/2026 3:24:06 AM
LDC01	1202	Error	DFSR	DFS Replication	6/2/2026 3:14:58 AM
LDC01	6104	Error	DFSR	DFS Replication	6/2/2026 3:14:57 AM

Type here to search

6:49 AM 6/2/2026

Server Manager

File and Storage Services > Servers

Active Directory Users and Computers

GG_IT_Admins Properties

General Members Member Of Managed By

Members:

Name	Active Directory Domain Services Folder
Sam Alonzo	lab.charles.com/Company_HQ/IT_Department

Add... Remove

OK Cancel Apply

LDC01 1202 Error DFSR DFS Replication 6/2/2026 3:24:06 AM

LDC01 1202 Error DFSR DFS Replication 6/2/2026 3:14:58 AM

LDC01 6104 Error DFSR DFS Replication 6/2/2026 3:14:57 AM

Type here to search

6:50 AM 6/2/2026

Server Manager

File and Storage Services > Servers

Active Directory Users and Computers

GG_IT_Admins Properties

General Members Member Of Managed By

Members:

Name	Type	Description
GG_IT_Admins	Security Group...	
Sam Alonzo	User	

Add... Remove

OK Cancel Apply

LDC01 1202

LDC01 1202

LDC01 6104

Type here to search

6:51 AM 6/2/2026

Automated Creation

```
#Create the HR Sub-OU inside existing Company_HQ container
New-ADOrganizationalUnit -Name "HR_Department" -Path
"OU=Company_HQ,DC=lab,DC=charles,DC=com" -ProtectedFromAccidentalDeletion $true

#Create the HR Security Group inside the new HR OU
New-ADGroup -Name "GG_HR_Staff" -GroupScope Global -GroupCategory Security -Path
"OU=HR_Department,OU=Company_HQ,DC=lab,DC=charles,DC=com"

#Create the HR Users inside the HR OU
$Password = ConvertTo-SecureString "SecureHR2026!" -AsPlainText -Force

New-ADUser -Name "Maria Santos" -GivenName "Maria" -Surname "Santos" `
  -SamAccountName "m_santos" -UserPrincipalName "m_santos@lab.charles.com" `
  -Path "OU=HR_Department,OU=Company_HQ,DC=lab,DC=charles,DC=com" `
  -AccountPassword $Password -Enabled $true

New-ADUser -Name "Ryan Reyes" -GivenName "Ryan" -Surname "Reyes" `
  -SamAccountName "r_reyes" -UserPrincipalName "r_reyes@lab.charles.com" `
  -Path "OU=HR_Department,OU=Company_HQ,DC=lab,DC=charles,DC=com" `
  -AccountPassword $Password -Enabled $true

#Add the new users into the HR Security Group
Add-ADGroupMember -Identity "GG_HR_Staff" -Members "m_santos", "r_reyes"

Write-Host "Successfully added"
```

Server Manager

Administrator: Windows PowerShell

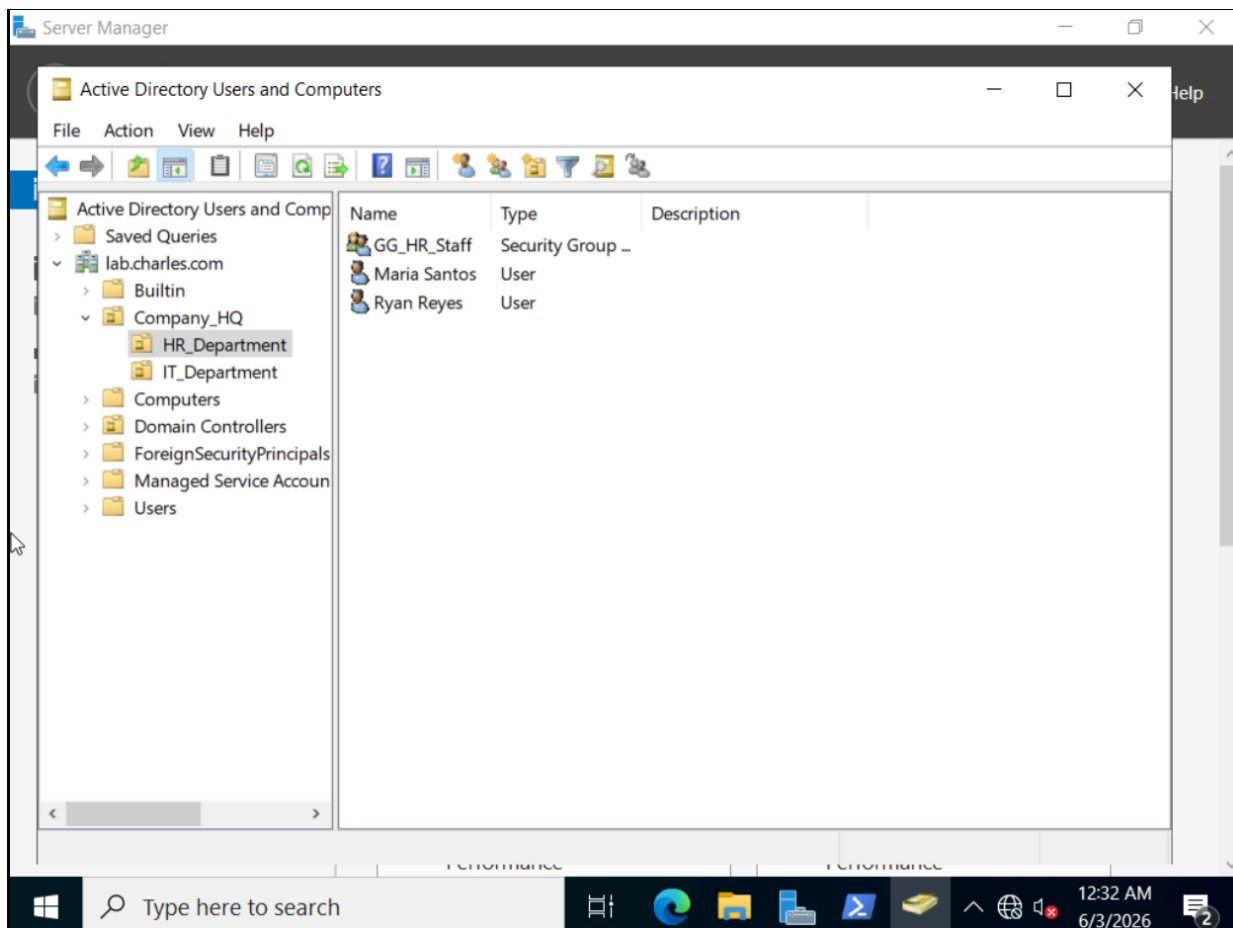
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

```
PS C:\Users\Administrator> New-ADOrganizationalUnit -Name "HR_Department" -Path "OU=Company_HQ,DC=lab,DC=char
les,DC=com" -ProtectedFromAccidentalDeletion $true
>>
>> New-ADGroup -Name "GG_HR_Staff" -GroupScope Global -GroupCategory Security -Path "OU=HR_Department,OU=Comp
any_HQ,DC=lab,DC=charles,DC=com"
>>
>> $Password = ConvertTo-SecureString "SecureHR2026!" -AsPlainText -Force
>>
>> New-ADUser -Name "Maria Santos" -GivenName "Maria" -Surname "Santos" `
>> -SamAccountName "m_santos" -UserPrincipalName "m_santos@lab.charles.com" `
>> -Path "OU=HR_Department,OU=Company_HQ,DC=lab,DC=charles,DC=com" `
>> -AccountPassword $Password -Enabled $true
>>
>> New-ADUser -Name "Ryan Reyes" -GivenName "Ryan" -Surname "Reyes" `
>> -SamAccountName "r_reyes" -UserPrincipalName "r_reyes@lab.charles.com" `
>> -Path "OU=HR_Department,OU=Company_HQ,DC=lab,DC=charles,DC=com" `
>> -AccountPassword $Password -Enabled $true
>>
>> Add-ADGroupMember -Identity "GG_HR_Staff" -Members "m_santos", "r_reyes"
>>
>> Write-Host "Successfully added"
Successfully added
PS C:\Users\Administrator>
```

Type here to search

12:31 AM
6/3/2026



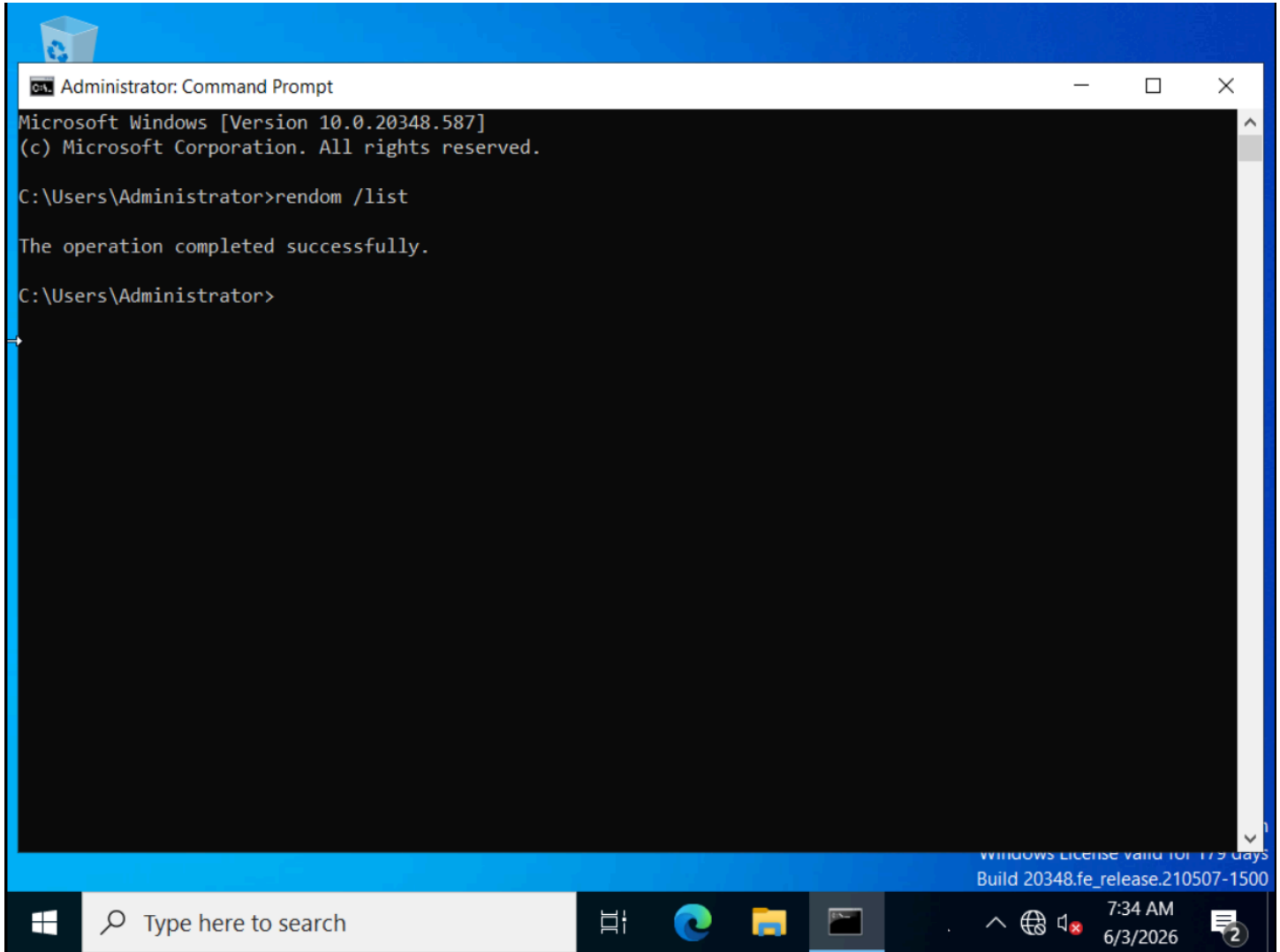
Renaming Domain

- Close aduc
- Open cdm (administrator)

Step 1: Generate architecture xml file

We need to export first our current domain blueprint in a text document that will be used later by using this command.

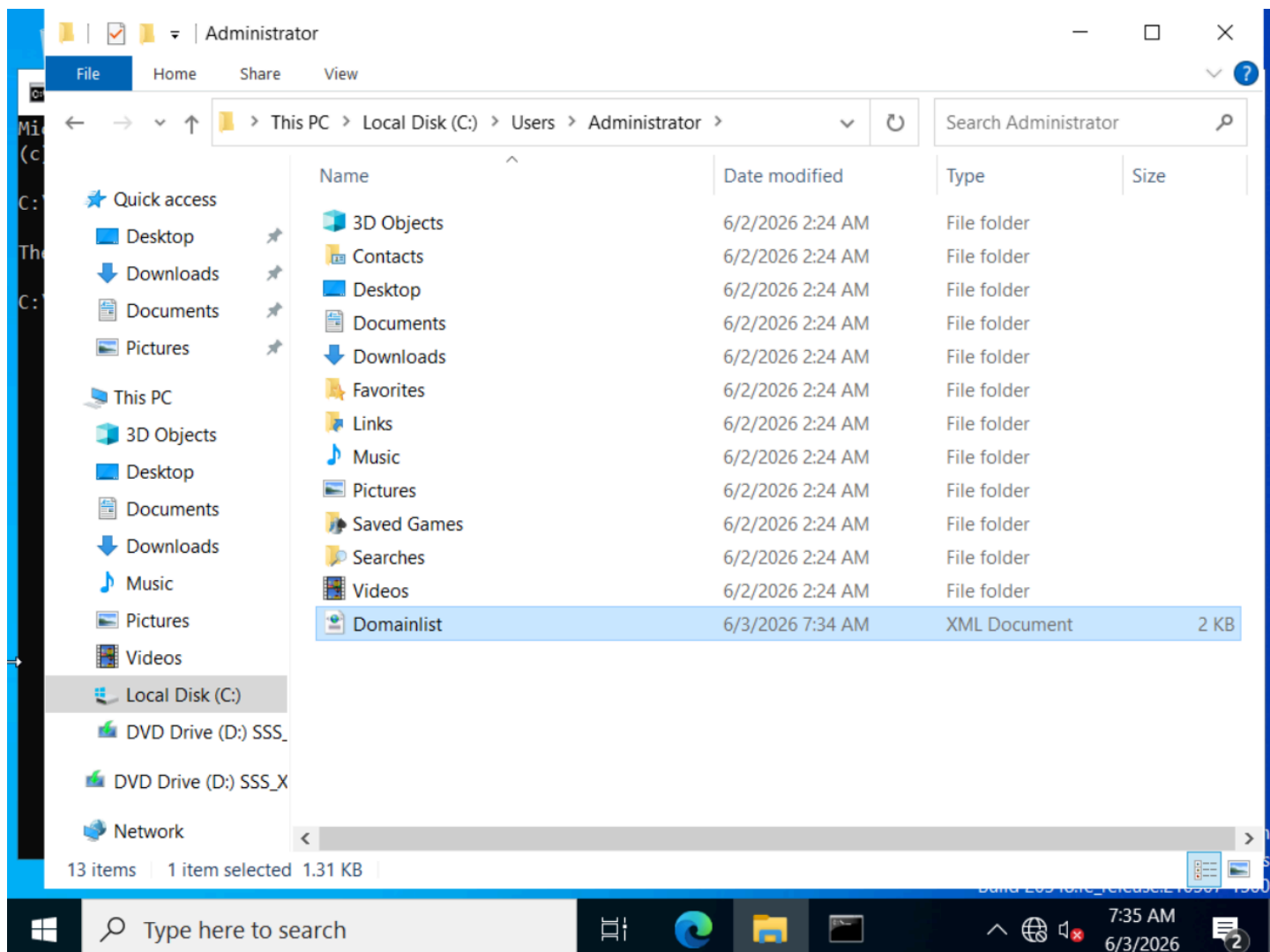
`random /list`



This will save a file named Domainlist.xml

Step 2: Edit the Domain Name in the XML

Open the XML file with Notepad



Replace Previous Domain names with new one and save the file

Domainlist - Notepad

File Edit Format View Help

```
<?xml version="1.0"?>
<Forest>
  <Domain>
    <!-- PartitionType:Application -->
    <Guid>b38177fe-a58a-4df0-bb6f-8989a8533fdb</Guid>
    <DNSname>ForestDnsZones.lab.charles.com</DNSname>
    <NetBiosName></NetBiosName>
    <DcName></DcName>
  </Domain>
  <Domain>
    <!-- PartitionType:Application -->
    <Guid>0c5ed86f-3456-4d82-a4fd-5c71d09b5f2f</Guid>
    <DNSname>DomainDnsZones.lab.charles.com</DNSname>
    <NetBiosName></NetBiosName>
    <DcName></DcName>
  </Domain>
  <Domain>
    <!-- ForestRoot -->
    <Guid>01e3f5ce-abce-47d5-ba0f-50ef7746e471</Guid>
    <DNSname>lab.charles.com</DNSname>
    <NetBiosName>LAB</NetBiosName>
    <DcName></DcName>
  </Domain>
</Forest>
```

Ln 1, Col 1 100% Windows (CRLF) UTF-16 LE

Type here to search

7:36 AM 6/3/2026

Domainlist - Notepad

File Edit Format View Help

```
<?xml version="1.0"?>
<Forest>
  <Domain>
    <!-- PartitionType:Application -->
    <Guid>01e3f5ce-abce-47d5-ba0f-50ef7746e471</Guid>
    <DNSName>lab.charles.com</DNSName>
    <NetBiosName>LAB</NetBiosName>
    <DcName></DcName>
  </Domain>
  <Domain>
    <!-- ForestRoot -->
    <Guid>01e3f5ce-abce-47d5-ba0f-50ef7746e471</Guid>
    <DNSName>lab.charles.com</DNSName>
    <NetBiosName>LAB</NetBiosName>
    <DcName></DcName>
  </Domain>
</Forest>
```

Replace

Find what: lab.charles.com

Replace with: lab.corp.local

☐ Match case

☐ Wrap around

Find Next

Replace

Replace All

Cancel

Ln 24, Col 10

100%

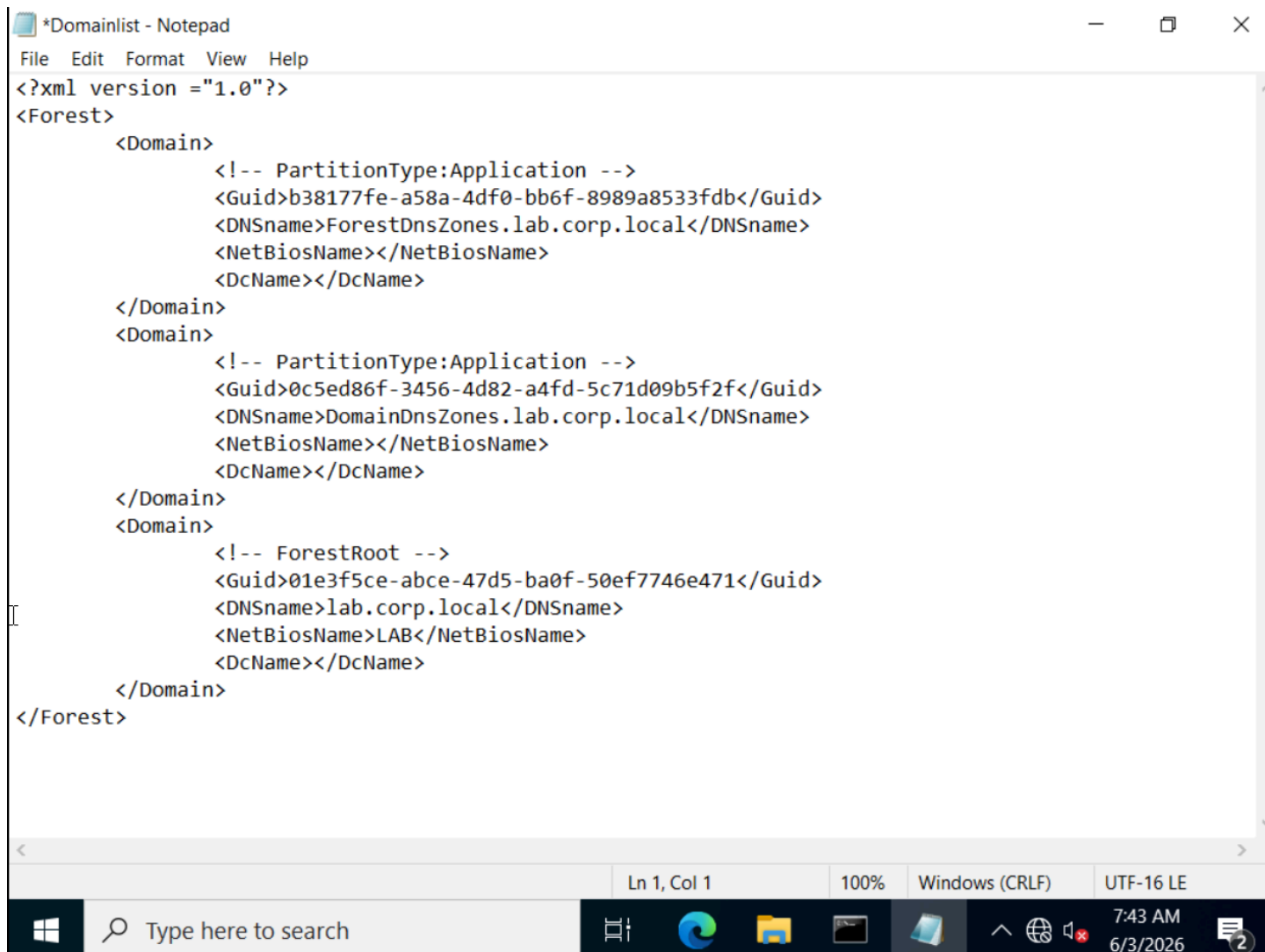
Windows (CRLF)

UTF-16 LE

Type here to search

7:38 AM

6/3/2026



```
*Domainlist - Notepad
File Edit Format View Help
<?xml version = "1.0"?>
<Forest>
  <Domain>
    <!-- PartitionType:Application -->
    <Guid>b38177fe-a58a-4df0-bb6f-8989a8533fdb</Guid>
    <DNSname>ForestDnsZones.lab.corp.local</DNSname>
    <NetBiosName></NetBiosName>
    <DcName></DcName>
  </Domain>
  <Domain>
    <!-- PartitionType:Application -->
    <Guid>0c5ed86f-3456-4d82-a4fd-5c71d09b5f2f</Guid>
    <DNSname>DomainDnsZones.lab.corp.local</DNSname>
    <NetBiosName></NetBiosName>
    <DcName></DcName>
  </Domain>
  <Domain>
    <!-- ForestRoot -->
    <Guid>01e3f5ce-abce-47d5-ba0f-50ef7746e471</Guid>
    <DNSname>lab.corp.local</DNSname>
    <NetBiosName>LAB</NetBiosName>
    <DcName></DcName>
  </Domain>
</Forest>
```

Note: Also update the <NetBiosName> if your changing root name

Step 3: Upload and Validate the new XML

rendom /upload (to parse the XML file into memory)

rendom /prepare (validate)

```
Domainlist - Notepad
File Edit Format View Help
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>random /list

The operation completed successfully.

C:\Users\Administrator>random /upload

The operation completed successfully.

C:\Users\Administrator>random /prepare
Waiting for DCs to reply.
Waiting for DCs to reply.
LDC01.lab.charles.com was prepared successfully
1 server contacted, 0 servers returned Errors
The operation completed successfully.

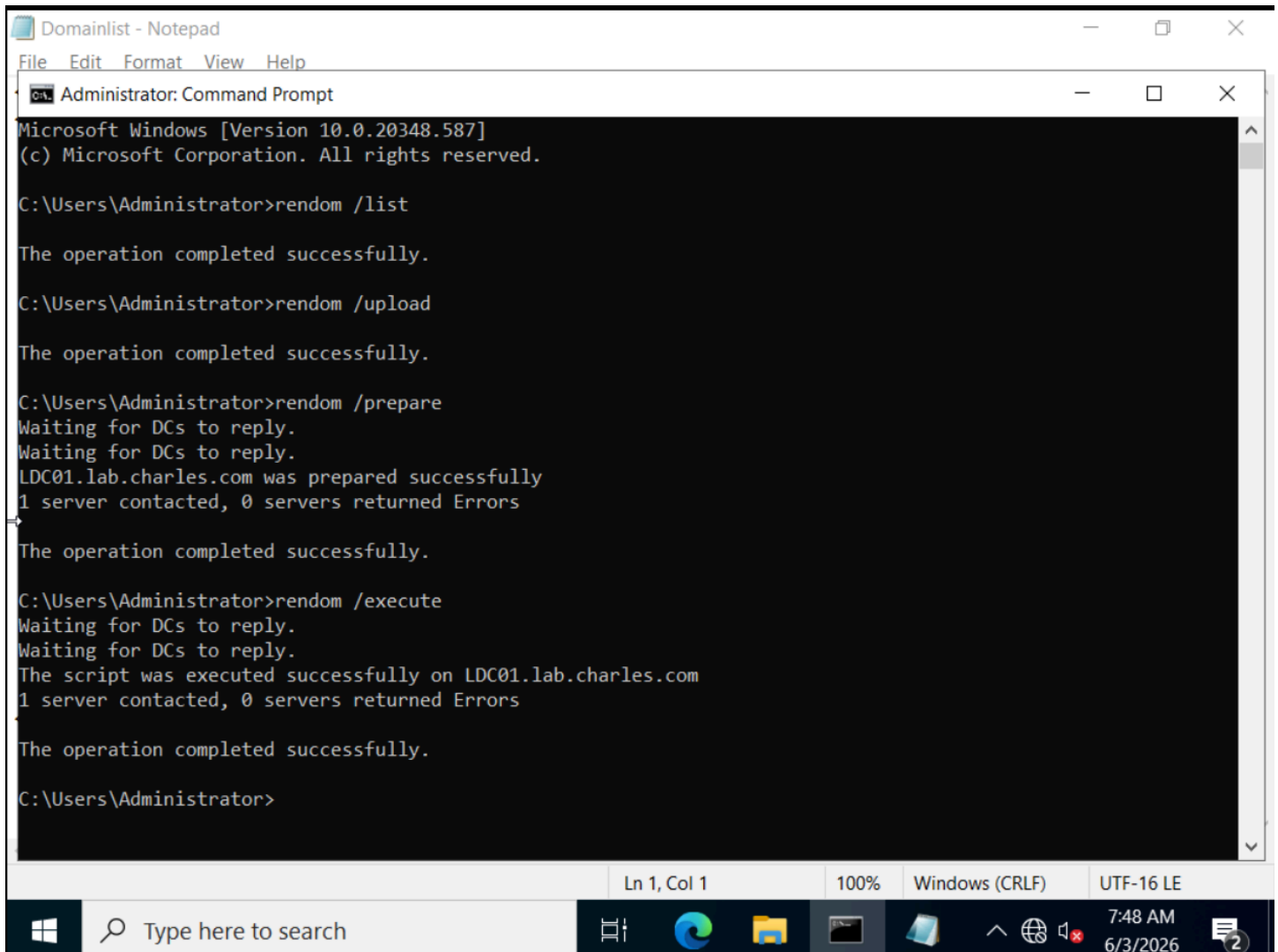
C:\Users\Administrator>
```

Ln 1, Col 1 100% Windows (CRLF) UTF-16 LE

Type here to search 7:47 AM 6/3/2026

Step 4: Execute the rename

random /execute (commits the change to the AD and triggers the sequence)



```
Domainlist - Notepad
File Edit Format View Help
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>random /list

The operation completed successfully.

C:\Users\Administrator>random /upload

The operation completed successfully.

C:\Users\Administrator>random /prepare
Waiting for DCs to reply.
Waiting for DCs to reply.
LDC01.lab.charles.com was prepared successfully
1 server contacted, 0 servers returned Errors

The operation completed successfully.

C:\Users\Administrator>random /execute
Waiting for DCs to reply.
Waiting for DCs to reply.
The script was executed successfully on LDC01.lab.charles.com
1 server contacted, 0 servers returned Errors

The operation completed successfully.

C:\Users\Administrator>
```

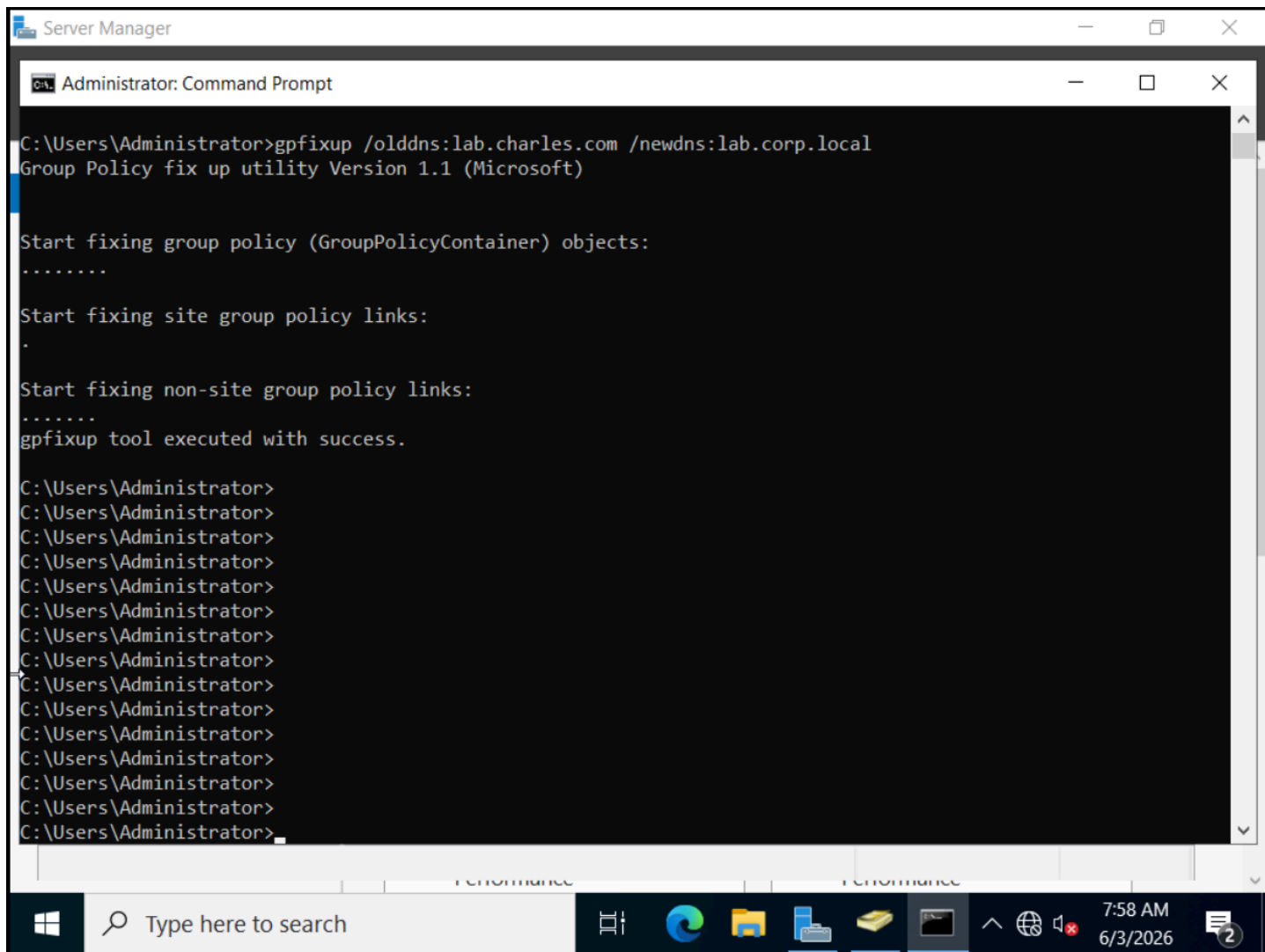
Note: The AD will reboot itself if successful

Step 5: Clean Up group policies

gpfixup /olddns:lab.charles.com /newdns:lab.corp.local.com

gpfixup /oldnb:(old) /newnb:(new) (we arent going to do it now since my dns was not changed)

random /end (clear temp data)



Server Manager

Active Directory Users and Computers

File Action View Help

Active Directory Users and Comp

- > Saved Queries
- > lab.corp.local

Name	Type	Description
Builtin	builtinDomain	
Company_HQ	Organizational ...	
Computers	Container	Default container for up...
Domain Con...	Organizational ...	Default container for do...
ForeignSecur...	Container	Default container for sec...
Managed Se...	Container	Default container for ma...
Users	Container	Default container for up...

Type here to search

7:59 AM
6/3/2026

3. Bulk User Provisioning & Password Compliance

A mock workforce database was parsed and imported into the directory structure to populate the organizational groups.

- Utilized an automated provisioning loop to generate individual accounts with unique User Principal Names (UPNs).
- Enforced standard default account rules: requiring a password change upon initial interactive desktop login and placing users directly into their designated department security groups.

[Screenshot Placeholder: [aduc_hierarchy_tree.png](#)]

Caption: Visual proof from Active Directory Users and Computers (ADUC) showing the structured OU tree and populated user objects.

Step 1: We need the CSV file to be used in provisioning, so create a csv file containing the data.

FirstName,LastName,Department,SamAccountName

Maria,Santos,HR,m_santos

Ryan,Reyes,HR,r_reyes

Juan,Dela Cruz,IT,j_delacruz

Step 2: Script for Parsing/Provisioning

1. Import the active directory module and the CSV file data

Import-Module ActiveDirectory

\$Employees = Import-Csv -Path "C:\LabFiles\mock_users.csv"

2. Establish a secure corporate default password baseline

\$DefaultPassword = ConvertTo-SecureString "WelcomeToCorp2026!" -AsPlainText -Force

3. Start the Automated Provisioning Loop

foreach (\$User in \$Employees) {

Construct paths dynamically based on the user's department column

```

$TargetOU = "OU= $($User.Department)_Department,OU=Company_HQ,DC=lab,DC=corp,DC=local"
$UPN = "$($User.SamAccountName)@lab.corp.local"
$FullName = "$($User.FirstName) $($User.LastName)"

# Create the user object with explicit compliance parameters
New-ADUser -Name $FullName `
    -GivenName $User.FirstName `
    -Surname $User.LastName `
    -SamAccountName $User.SamAccountName `
    -UserPrincipalName $UPN `
    -Path $TargetOU `
    -AccountPassword $DefaultPassword `
    -ChangePasswordAtLogon $true `
    -Enabled $true

# Instantly place the account into its functional security group
$TargetGroup = "GG_ $($User.Department)_Staff"
Add-ADGroupMember -Identity $TargetGroup -Members $User.SamAccountName
}

```

`Import-Csv` loads the employee spreadsheet while `ConvertTo-SecureString` converts the plain text password into an encrypted system object. Inside the `foreach` loop, the script reads each row to dynamically build unique folder paths via `$TargetOU` and network login IDs using `$UPN`. Finally, `New-ADUser` creates the account with `-ChangePasswordAtLogon $true` to enforce the initial compliance reset, and `Add-ADGroupMember` instantly maps the user to their department group.

Step 3; Run the script

◆ Verification & Testing

Test 1: DNS Resolution & Domain Health Audit

A healthy directory infrastructure relies completely on solid DNS records. Verification steps were run locally to ensure proper name resolution and record creation.

- **Command Executed:** `nslookup homelab.local` and `dcdiag /test:Connectivity`
- **Expected Behavior:** The local DNS server must resolve the domain to the DC's static IP address without dropping packets.
- **Result Verification:**

DOS

- C:\Users\Administrator> nslookup homelab.local
- Server: UnKnown
- Address: 127.0.0.1
-
- Name: homelab.local
- Addresses: 192.168.10.10

Test 2: Directory Object Verification via CLI

To prove the directory object parameters were successfully mapped and locked down during the bulk automation phase, an object query was executed using the Active Directory administrative CLI module.

- **Command Executed:** `Get-ADUser -Identity "charl_admin" -Properties MemberOf, PasswordExpired`
- **Expected Behavior:** The account properties should confirm its exact group membership and reflect security parameter flags.
- **Result Telemetry:**

Plaintext

- DistinguishedName : CN=Charl Admin,OU=IT,OU=Departments,DC=homelab,DC=local
- Enabled : True
- MemberOf : {CN=GG_IT_Admins,OU=Groups,DC=homelab,DC=local}
- PasswordExpired : True (Forces password reset on first login simulation)
- ``
-
- ---
-
- ## ♦ Key Takeaways & Enterprise Security Impact
- * **Scalability via Code:** By relying on automated PowerShell provisioning arrays instead of manual GUI point-and-clicking, setup deployment times were reduced by over 90%, proving production readiness.
- * **Access Control Baseline:** The structure separates standard departmental tier access from enterprise domain administration, creating a foundational layout that eliminates arbitrary nested privileges and restricts systemic lateral movement options for potential threats.



Pro-Tip for Project 1 Deliverables:

When you write your actual scripts (like `Create-OUTs.ps1`), **never** hardcode an actual domain administrator password inside the plaintext file. If a recruiter checks your GitHub repo and sees a real password written out in your script, it's an immediate red flag.

Instead, use standard secure inputs like `Read-Host -AsSecureString` to show them you already know standard secure coding and defensive infrastructure habits!